

Programme	B. Sc. Mathematics Honours			
Course Code	MAT3MN201			
Course Title	CALCULUS OF SEVERAL VARIABLES			
Type of Course	Minor			
Semester	III			
Academic Level	200 - 299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Calculus of Single Variable			
Course Summary	This course provides a comprehensive study of advanced calculus topics, including partial derivatives, limits, continuity, the chain rule, and vector-valued functions. Students will explore directional derivatives, tangent planes, and extrema of functions of multiple variables, as well as integral calculus techniques such as line integrals, double integrals (including those in polar coordinates), surface integrals, and the applications of these concepts in vector calculus and field theory			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply Multivariable Calculus Concepts to Vector Valued Functions	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply Techniques of Multivariable Integration	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva / End Sem Exam
CO3	Apply Advanced Theorems in Multivariable Calculus	E	C	Internal Exam/ Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text book	Calculus, Soo T. Tan, Brooks/Cole Cengage Learning (2010) ISBN-13: 978-0-534-46579-7			
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I	Partial Derivatives		14	17
	1	12.1: Vector Valued Functions & Space Curves		
	2	12.2: Differentiation & Integration of Vector Valued Functions		
	3	13.1: Functions of Two or More Variables (Computer Graphics and Sketching of Contour Maps are optional)		
	4	13.2: Limits & Continuity (Problems asking about the continuity of function are optional, as is the $\epsilon - \delta$ definition)		
	5	13.3: Partial Derivatives (Examples 4, 5 and 6 are optional)		
	6	13.4: Differentials (Examples 4 & 6 and Error in Approximating ΔZ are optional topics)		
	7	13.5: The Chain Rule (Example 2 is optional)		
	8	13.6: Directional Derivatives (Examples 6 and 7 are optional)		
	9	13.7: Tangent Planes & Normal Lines (Use of tangent planes in approximating surfaces is an optional topic)		
	10	13.8: Extrema of Functions of Two Variables (This section is not intended for external examination)		
II	Vector Derivatives – Calculus of Scalar & Vector Fields		11	18
	11	15.1 : Vector Fields (Examples 3 and 4 are optional)		
	12	15.2: Divergence & Curl		
	13	15.3: Line Integrals		
	14	15.4: Independence of Path & Conservative Vector Fields (subsection Conservation of Energy is optional)		
III	Multiple Integration		14	18
	15	14.1: Double Integrals (Examples 2 and 3 are optional)		
	16	14.2: Iterated Integrals		

	17	14.3: Double Integrals in Polar Coordinates		
	18	14.4: Applications of Double Integrals (This section is not intended for external examination)		
	19	14.5: Surface Area (Last subsection on Area of surface given by two equations is optional)		
	20	14.6: Triple Integrals (up to and including Example 3)		
	21	14.7: Triple Integrals in Cylindrical & Spherical Coordinates		
	22	14.8: Change of Variables in Multiple Integrals (This section is not intended for external examination)		
IV	Integral Calculus of Fields & Fundamental Theorems		11	17
	23	15.5: Green's Theorem (up to and including Example 3)		
	24	15.6: Parametric Surfaces (This section is not intended for external examination)		
	25	15.7: Surface Integrals (up to and including Example 3)		
	26	15.8: Divergence Theorem (up to and including Example 3)		
	27	15.9: Stoke's Theorem (up to and including Example 3)		
V	Open Ended Module – Complex Analysis		12	
	1	Algebra of Complex Numbers, Complex Functions, Complex Differentiation		
	2	Cauchy-Riemann Equations, Analytic Functions		
	3	Complex Line Integrals		
	4	Cauchy's & Cauchy-Goursat Theorems		
	5	Cauchy's Integral Formula, Derivative Formula		
	6	Morera's & Liouville's Theorem, Fundamental Theorem of Algebra		
	7	12.3: Arc Length & Curvature		
	8	12.4: Velocity & Acceleration		
	9	12.5: Tangential & Normal Components		
	10	13.9: Lagrange Multipliers		

References:

1. Advanced Engineering Mathematics, Erwin Kreyzsig, 10th Edition, Wiley India.
2. Advanced Engineering Mathematics, 6th Edition, Dennis G. Zill, Jones & Bartlett Learning LLC (2018) ISBN: 978-1-284-10590-2.
3. Calculus & Analytic Geometry, 9th Edition, George B. Thomas & Ross L. Finney, Pearson Publications.
4. Thomas' Calculus, 14th Edition, Maurice D. Weir, Christopher Heil, & Joel Hass, Pearson Publications.
5. Calculus, 7th Edition, Howard Anton, Biven, & Stephen Davis, Wiley India.

Note: 1) Optional topics are exempted for end semester examination.

2) Proofs of all the results are also exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	2	3	3	3	1	2
CO 2	3	0	3	2	3	3	3	1	2
CO 3	3	2	3	2	3	3	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓